Neural network representation of Boolean circuits

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Most standard 2-bit gates can be represented by a single layer perceptron (SLP). These include the `AND`, `OR`, `NAND`, and `NOR` gates. This can be understood geometrically by considering the 2-bit dataset $S \equiv \{(0,0),(0,1),(1,0),(1,1)\}$ to be coordinates of the corners of a square. These four gates classify S in such a way that one could represent it by drawing a single boundary-line through the square.

By contrast, the classification performed by `XOR` will require two boundaries. Consequently , `XOR` cannot be represented by an SLP and the minimal neural network requires one hidden layer with two neurons—an 2-2-1 multi-layer perceptron (MLP).